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PATENT INFORMATION EXTRACTION FORM

TOPIC : CyberGravity Memory Architecture (CGMA): A Physics-Inspired AI Framework for Dynamic Cyber Trust Propagation and Adaptive Memory Evolution

Suggested expanded title for broader patent scope: “A System and Method for Modelling Inter-Process Security Trust as a Gravitational Field and Coupling Threat-Memory Decay, Reinforcement, and Archival to that Field for Autonomous Endpoint Defence”

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Answer the following questions in brief: (use extra sheets if needed, submit pics/videos etc. which can help us to understand the invention better)

1. What was the problem?

Two things go wrong in today's EDR systems, and nobody has connected them. First, trust between processes (Chrome spawning PowerShell, PowerShell touching the Registry, a Registry write before an LSASS access) is scored one hop at a time. Each hop can look mild on its own, so a slow, staged attack walks right through – nothing ever crosses a threshold. Second, old attack memory is aged out purely by the clock: it fades regardless of whether the technique behind it is still active anywhere in the environment. So a still-relevant memory gets forgotten just because time passed, and stale noise sticks around just because it's recent. No system today lets the live pull a process chain is currently exerting decide both things at once – whether to act on that chain, and how fast the memories tied to it should fade, get reinforced, or get archived.

2. How did you solve it (Inventive Step)?

CyberGravity Memory Architecture (CGMA) treats every running process as a mass in a security field, and treats trust between two processes as a gravitational pull rather than a fixed, independent score. A process's pull comes from how privileged it is, how much its code and origin can be trusted, how many other processes it spawns, and how sensitive the resources it can reach are. Two processes pull on each other more strongly the more mass they carry and the less they trust one another, and that pull weakens the further apart they sit in the process lineage.

The key departure from ordinary risk scoring is that this pull is *additive along a chain* rather than judged hop by hop. A four-step chain such as Chrome spawning PowerShell, PowerShell touching the Registry, and that Registry write reaching toward LSASS builds up cumulative pull across the whole path, even though each individual hop looks unremarkable in isolation. Once that accumulated pull crosses a calibrated threshold, the region is treated as a "gravity well" – a chain shaped like an attack even if no single link in it would ever trip a conventional per-hop alert. This is the non-obvious part: existing lineage-graph and EDR scoring systems evaluate edges independently, so they structurally cannot see a threat that only becomes visible when pull is summed across several hops at once.

Once a well is detected, the system doesn't jump straight to killing the process. It works out how much counter-pressure would be needed to pull the chain back to safety, and ranks available responses – a re-authentication challenge, a handle revocation, a suspend – by how cheaply each one clears that counter-pressure, trying the least disruptive option first. Only if the pull keeps building past a second, higher threshold does the system act on its own and contain the process outright, because past that point no lighter response could realistically catch up before the chain reaches its target.

The second half of the invention is what happens to memory of past attacks, and this is where the real coupling sits. Instead of aging every stored attack memory on a shared clock, CGMA ties each memory's decay to how much pull its associated technique is *still exerting somewhere in the live environment right now*. A memory tied to a technique that is still actively showing up decays slowly and gets reinforced; a memory tied to a technique with zero current relevance fades quickly no matter how recently it was recorded. This is the specific technical mechanism that separates CGMA from every existing confidence-decay or timestamp-decay memory system: the decay rate is not a schedule, it is a live output of the same gravitational field used for real-time defence – one computation drives both what the system does right now and what it chooses to remember. That coupling, not the physics language used to describe it, is the inventive step.

Memories are continuously sorted into three states purely by how much relevance-weight they still carry: reinforced (kept hot and fed back into future pull calculations, making related live processes easier to flag), archived (moved to cold, sealed storage but still retrievable), or collapsed (compressed down to a bare signature rather than deleted outright). If a future live event matches a collapsed signature, the memory is pulled back into active use immediately – nothing here requires retraining any underlying detection model, since reinforcing, archiving, and collapsing all happen at the level of the memory's relevance-weight, not at the level of model weights.

3. What were the other possible solutions and why they could not be done?

Static/graph-weighted trust scoring (process-lineage EDR graphs, ATT&CK weighted scoring): judges each edge on its own, so it cannot accumulate pull across a chain the way CGMA does – a quiet multi-hop path stays invisible until it's already too late.

Timestamp/confidence memory decay (existing memory-graph and adaptive-memory tools): ages every stored memory on the same clock, conflating recency with relevance – operationally useful old memories vanish while useless recent noise sticks around.

Adaptive security posture systems (the class of art represented by patents such as US12047400B2): shift the whole system between broad postures like strict or relaxed, but posture is one global mode, not a continuous per-process, per-relationship field, and none of this art couples posture to how memory ages at all.

A version that simply renamed an existing risk score as "gravity" was considered and rejected early in the design – a label is not a mechanism, and would not stand up as inventive. What makes CGMA both novel and non-obvious is that no prior system lets a live, chain-accumulated trust field directly govern the aging rate of a separate memory store; each piece of prior art solves either the detection half or the memory half, never both from one shared computation, and nothing in that prior art would have suggested tying the two together this way.

4. What are the advantages of the solution proposed by you?

- **Chain-level detection:** cumulative pull across a full lineage catches multi-hop attacks that per-hop scoring misses.
- **Proportionate response:** the system always tries the least disruptive action that still neutralises the threat before escalating.
- **Safety margin on irreversible actions:** automatic containment fires before a chain can reach its target, not after.
- **Memory tied to relevance, not recency:** live-relevant attack knowledge stays reinforced while dead noise fades fast.
- **Forgetting is reversible:** collapsed memories are compressed, not deleted, and can be recovered the moment they become relevant again.
- **No retraining needed:** reinforcement, archiving, and collapse all happen at the memory-relevance level, never touching the underlying detection models.

5. Explain the stepwise working of the innovation explaining all the components used in the invention and the specific function they are performing.

Phase 1: Process Graph Ingestion and Mass Assignment

The Process Telemetry Collector pulls live process, handle, and registry activity into a lineage graph. The Mass Estimator scores each process on privilege, code/provenance trust, how many children it spawns, and how sensitive what it can reach is. The Trust Propagation Engine sets an initial trust value on every parent–child relationship using history held by the memory subsystem (Phase 4).

Phase 2: Gravitational Field Construction

The Gravitational Force Computation Unit scores every live relationship in the graph. The Security Gravity Field Map sums that pull along each full lineage path, continuously, as the graph changes in real time.

Phase 3: Well Detection and Graduated Response

The Gravity Well Detector flags any subgraph whose accumulated pull passes the well threshold. The Escape Velocity Threshold Module works out the cheapest defensive action that still clears the danger, and applies it. If pull keeps rising past a second, higher threshold, the Event Horizon Response Engine contains the process outright, with no further confirmation needed.

Phase 4: Attack Memory Encoding

Every contained or confirmed event – its technique fingerprint, its lineage snapshot, and the pull recorded at the time – is logged by the Attack Memory Encoder with an initial relevance-weight.

Phase 5: Gravity-Coupled Memory Aging

The Memory Orbit Calculator continuously measures how much pull each stored event's technique is still exerting somewhere in the live field. The Decay Function Engine updates every memory's relevance-weight each cycle, decaying it faster when that live pull is low and slower when it is high.

Phase 6: Reinforcement, Archival, and Collapse

The Memory Reinforcement & Archival Classifier sorts each memory by its current relevance-weight into reinforced (fed back into Phase 2's pull calculations), archived (cold, sealed, retrievable), or collapsed (compressed to a signature by the Knowledge Collapse & Reconstruction Module). A later live event matching a collapsed signature pulls that memory back into active use immediately, feeding straight back into Phase 1's trust values for the next cycle – closing the loop without retraining any underlying model.

6. Attach drawing hand-made/ computer made showing all the components of the Invention.

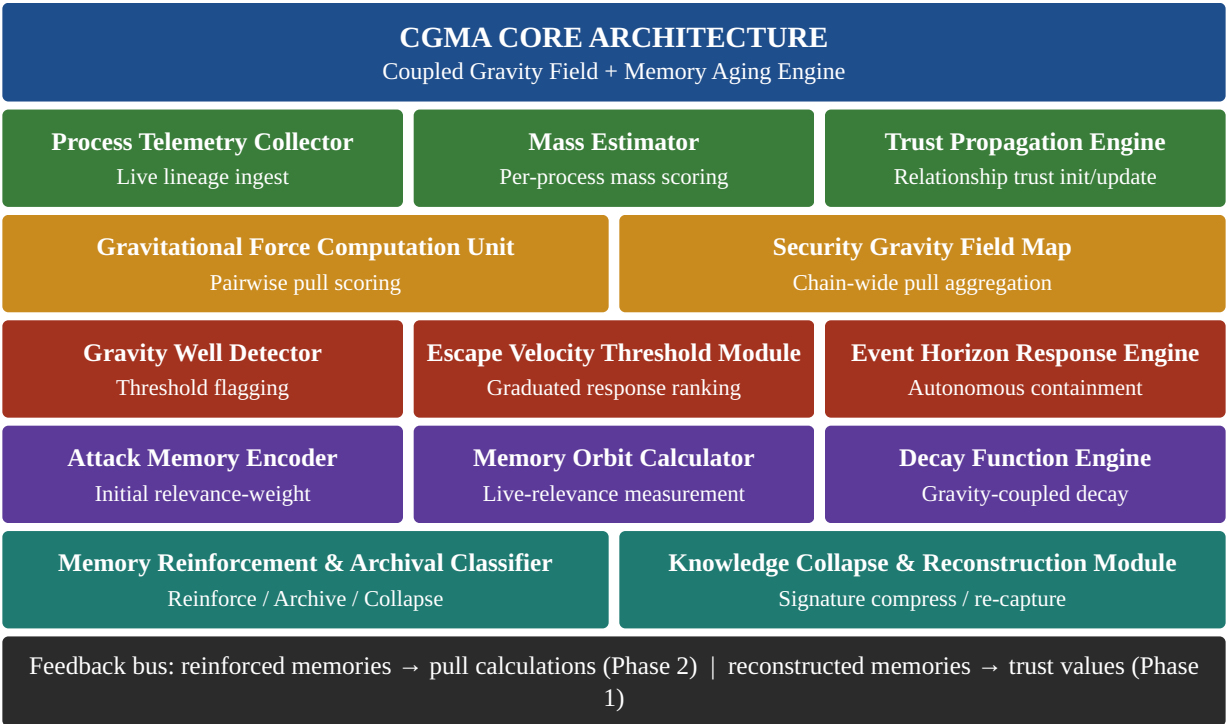


Figure 1: CGMA Core Architecture — five coupled subsystems: lineage ingestion, gravity field construction, well detection & response, memory encoding, and gravity-coupled aging, closed by a reinforcement feedback bus.

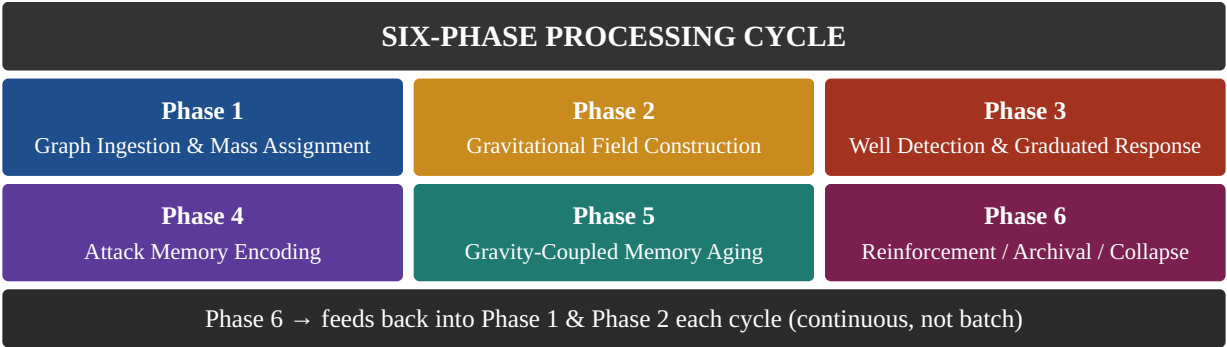


Figure 2: Six-Phase Processing Cycle — each phase's output continuously re-enters the earlier phases, so the gravity field and the memory store co-evolve every telemetry cycle.

Relevant Key phrases relating to your invention

S.NO. Key phrases

1. Physics-inspired security trust field / gravitational trust propagation
2. Additive multi-hop process-lineage pull accumulation
3. Gravity well detection for living-off-the-land attack chains
4. Escape-velocity-ranked graduated defensive response
5. Event-horizon autonomous containment threshold
6. Gravity-coupled attack memory decay driven by live relevance
7. Live-relevance measurement for stored attack memories
8. Reinforcement / Archival / Collapse memory-state classifier
9. Knowledge collapse and reconstruction without model retraining
10. Relevance-driven (non-timestamp) threat memory aging
11. Process mass – trust – memory density composite scoring
12. Memory-relevance reinforcement decoupled from model weights